

Amshavarthana Sreesivanathasarma

Software Engineering Undergraduate | Aspiring Data Science & Machine Learning Intern

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🌐 LinkedIn | 🐙 GitHub | 📝 Medium | 🌐 Portfolio

Professional Summary

Software Engineering undergraduate at the University of Kelaniya (GPA 3.71/4.00) with a strong interest in Data Science and Machine Learning. Skilled in Python, Data Analysis, and building end-to-end ML web applications and AI-powered tools. Committed to clean code and solving real-world problems analytically.

Education

- **University of Kelaniya** 2024 – Present
BSc (Hons) in Software Engineering **Current GPA: 3.71 / 4.00**
Specialized in Data Science, Net Centric Application Development, and Business Engineering Applications.
- **J/Arunodaya College** 2020 – 2023
G.C.E. Advanced Level Examination (2023) Combined Mathematics, Chemistry, Physics **Z-Score: 1.5368**

Projects

- **Flight Fare Prediction System** *Individual* [Source Code](#)
Python, Pandas, NumPy, Scikit-learn, Flask, HTML
 - Performed EDA and feature engineering on historical flight data (airline, route, duration, stops) to prepare inputs for ML modelling.
 - Developed a Random Forest Regressor achieving an R-squared of 0.81 and MAPE of 13.3% on ticket fare prediction.
 - Deployed the model via a Flask web interface for real-time fare estimation.
- **EduMind AI** *Individual* [Live Demo](#) [Source Code](#)
Python, Streamlit, Groq LLaMA, scikit-learn, XGBoost, SQLite, Plotly
 - Built a trilingual (English, Tamil, Sinhala) AI study companion for Sri Lankan A/L and university students, powered by Groq LLaMA.
 - Built and evaluated Random Forest and XGBoost classifiers on 6,389 real student records across 5 datasets, achieving 93.85% accuracy for academic risk prediction and 85.31% for dropout detection.
 - Integrated trained models into the live Streamlit app for real-time student risk assessment, visualized with an interactive Plotly heatmap backed by SQLite.
- **Alexa Voice Assistant** *Individual* [Source Code](#)
Python, Streamlit, OpenAI Whisper, scikit-learn MLP, TF-IDF, SoundDevice, FFmpeg
 - Built real-time speech-to-text using OpenAI Whisper with low-latency response handling.
 - Engineered a TF-IDF + MLP intent-detection pipeline to classify and route 20+ programmable voice commands (web search, app launch, system control).
 - Designed a dark glassmorphic Streamlit UI with TTS audio output.

Technical Skills

- **Languages:** Python, SQL, Java, JavaScript
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, XGBoost, TF-IDF, MLP Neural Networks
- **Frameworks:** Flask, Spring Boot, Streamlit, OpenAI Whisper, Groq API
- **Databases:** MySQL, SQLite
- **Visualization:** Matplotlib, Seaborn, Plotly, Power BI

Certifications

- **The Complete Python Masterclass** – Udemy [View Certificate](#)
- **Statistics Foundations** – LinkedIn Learning [View Certificate](#)
- **Data Analytics Course** – LinkedIn Learning [View Certificate](#)

Soft Skills & Languages

Soft Skills: Team Collaboration, Time Management, Critical Thinking, Problem Solving, Adaptability & Quick Learning

Languages: Tamil (Native), English (Fluent), Sinhala (Basic)

References

- **Prof. (Mrs.) I. U. Hewapathirana**, Senior Lecturer, Faculty of Science, University of Kelaniya
✉ ihewapathirana@kln.ac.lk
- **Dr. Nalin Warnajith**, Senior Lecturer, Faculty of Science, University of Kelaniya
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